

Gout and Hyperuricemia

AAU
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Definition

Gout is a spectrum of disorders including hyperuricemia, recurrent acute arthritis caused by monosodium urate crystals in synovial fluid, tophi deposition, interstitial renal disease, and uric acid nephrolithiasis.

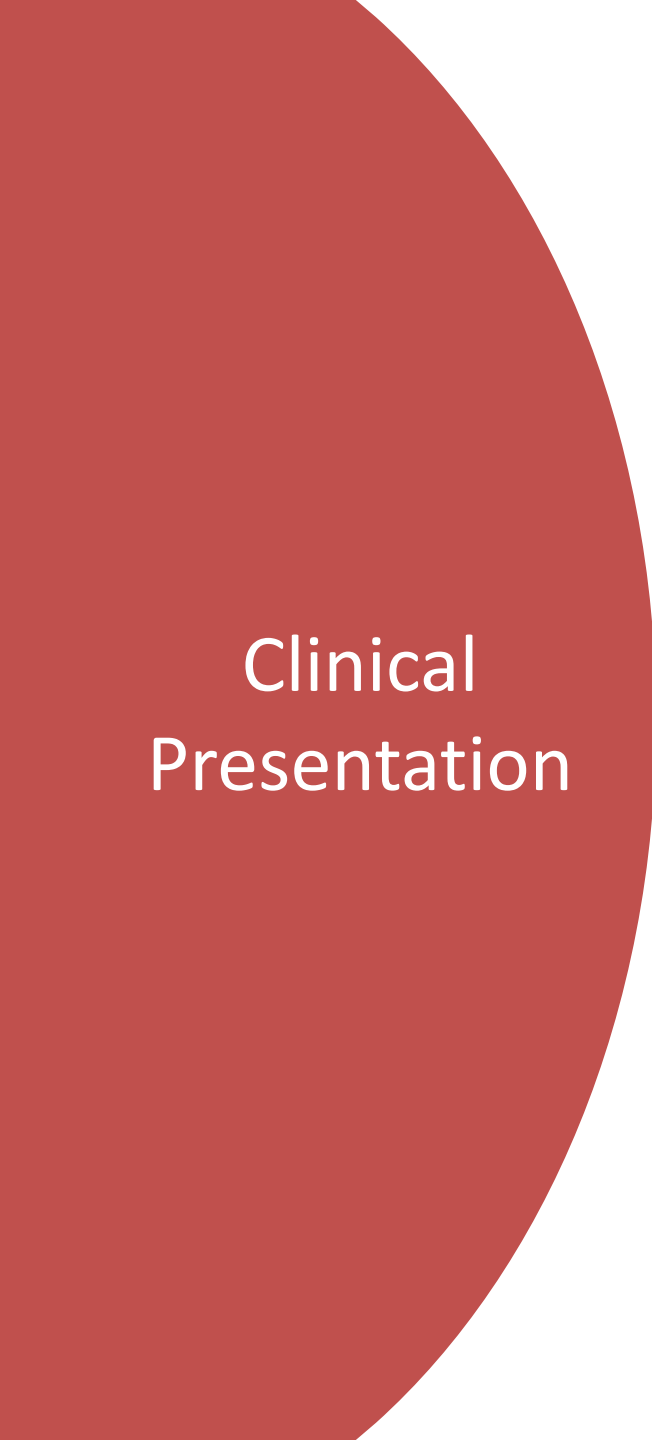
Hyperuricemia is defined as a serum urate concentration greater than 7.0 mg/dL.

Pathophysiology

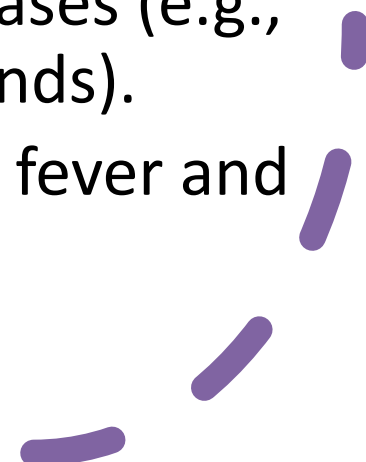
- Uric acid is the end product of purine degradation.
- Overproduction may occur due to enzyme abnormalities (PRPP synthetase hyperactivity, HGPRT deficiency).
- Underexcretion results from decreased renal clearance (diuretics, ethanol, cyclosporine, etc.).
- Deposition of urate crystals in joints leads to inflammation, pain, and tissue damage.

PRPP synthetase (phosphoribosyl pyrophosphate synthetase)

HGPRT (hypoxanthine-guanine phosphoribosyltransferase)



Clinical Presentation

- Acute gouty arthritis: sudden, severe pain, redness, swelling, warmth.
 - Most commonly affects the first metatarsophalangeal joint (podagra).
 - May involve instep, ankles, knees, wrists, or elbows.
 - Tophi form in chronic cases (e.g., ear, Achilles tendon, hands).
 - Can be associated with fever and leukocytosis.
- 



Tophi lumps



Diagnosis

- Definitive: synovial fluid aspiration showing monosodium urate crystals.
- Supportive: typical clinical features and therapeutic response.
- Serum uric acid >7 mg/dL supports, but does not confirm, diagnosis.

Treatment Goals

- Terminate acute attack.

- Prevent recurrent gouty arthritis.

- Prevent complications (tophi, nephrolithiasis, nephropathy).

Nonpharmacologic Therapy

- Reduce dietary purines (organ meats, seafood).
- Avoid alcohol and high-fructose drinks.
- Maintain hydration and healthy weight.
- Joint rest and local ice during acute attacks.

Acute Gouty Arthritis Management

- 1. NSAIDs (first-line): indomethacin, naproxen, sulindac.
 - High dose until pain relief, then taper.
- 2. Colchicine: 1 mg initially, then 0.5 mg every 1 hour (max 8 mg/day).
- 3. Corticosteroids (oral, parenteral, intraarticular) for NSAID/colchicine intolerance.
- 4. Analgesics for pain management.
- → Acute attack resolves within 5–8 days.

Prophylaxis and Maintenance Therapy

- Indicated for recurrent attacks (>2/year) or tophi.

- Colchicine 0.5–0.6 mg once or twice daily for prevention.

- Maintain serum uric acid <6 mg/dL (target <5 mg/dL if tophaceous gout).

Uric Acid–Lowering Therapy



Allopurinol: Xanthine oxidase inhibitor; 100–300 mg/day (up to 800 mg/day).

Reduce dose in renal impairment.
Adverse effects: rash, hypersensitivity, GI upset.



Probenecid /
Sulfinpyrazone: Uricosurics for underexcretors.

Promote uric acid excretion.
Maintain alkaline urine to avoid stones.



Febuxostat: Alternative xanthine oxidase inhibitor for allopurinol intolerance.

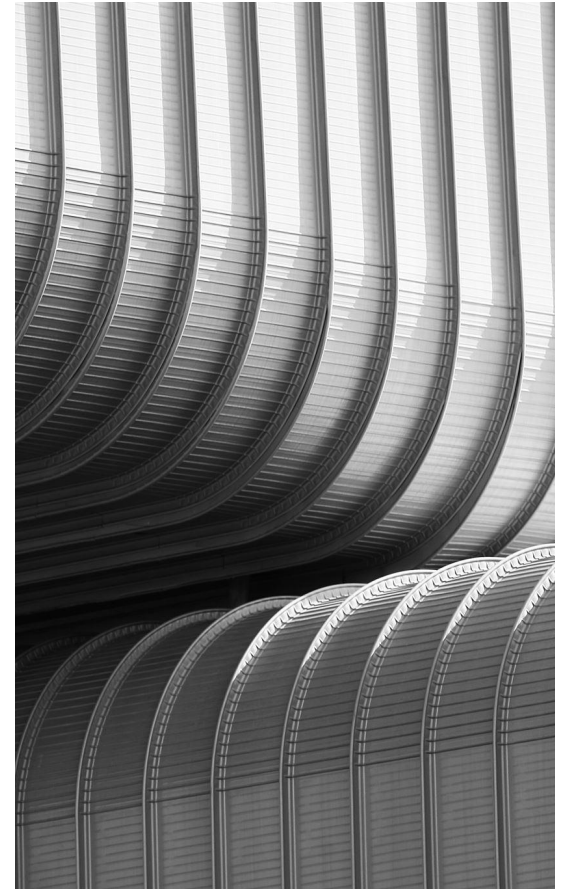
Evaluation of Therapeutic Outcomes

- Relief of pain within 8–12 hours; full resolution in 48–72 hours.

- Monitor serum uric acid periodically.

- Assess renal and hepatic function every 6–12 months.

- Monitor for adverse effects (GI upset, rash, cytopenia).



Reading assignment

- Suggested readings:
 - Diagram: Purine metabolism and uric acid formation.
 - Flowchart: Acute gout management algorithm (NSAIDs → Colchicine → Corticosteroids).
 - Microscopic image: Needle-shaped urate crystals.
 - X-ray: Joint erosion and tophi.

References

- Pharmacotherapy Handbook, 7th Edition – Chapter 1: Gout and Hyperuricemia.
- American College of Rheumatology (ACR) Guidelines.
- Recent clinical updates on gout management.